Hack-A-Bot 2026
Project 6: Grid Box

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Problem & Our Solution

30%
of factory energy
is wasted

Motors run at full speed
when partial load is enough

£180B/year
lost to undetected
equipment faults

4+ hours average
downtime per fault

£162,000+
cost of industrial
SCADA system

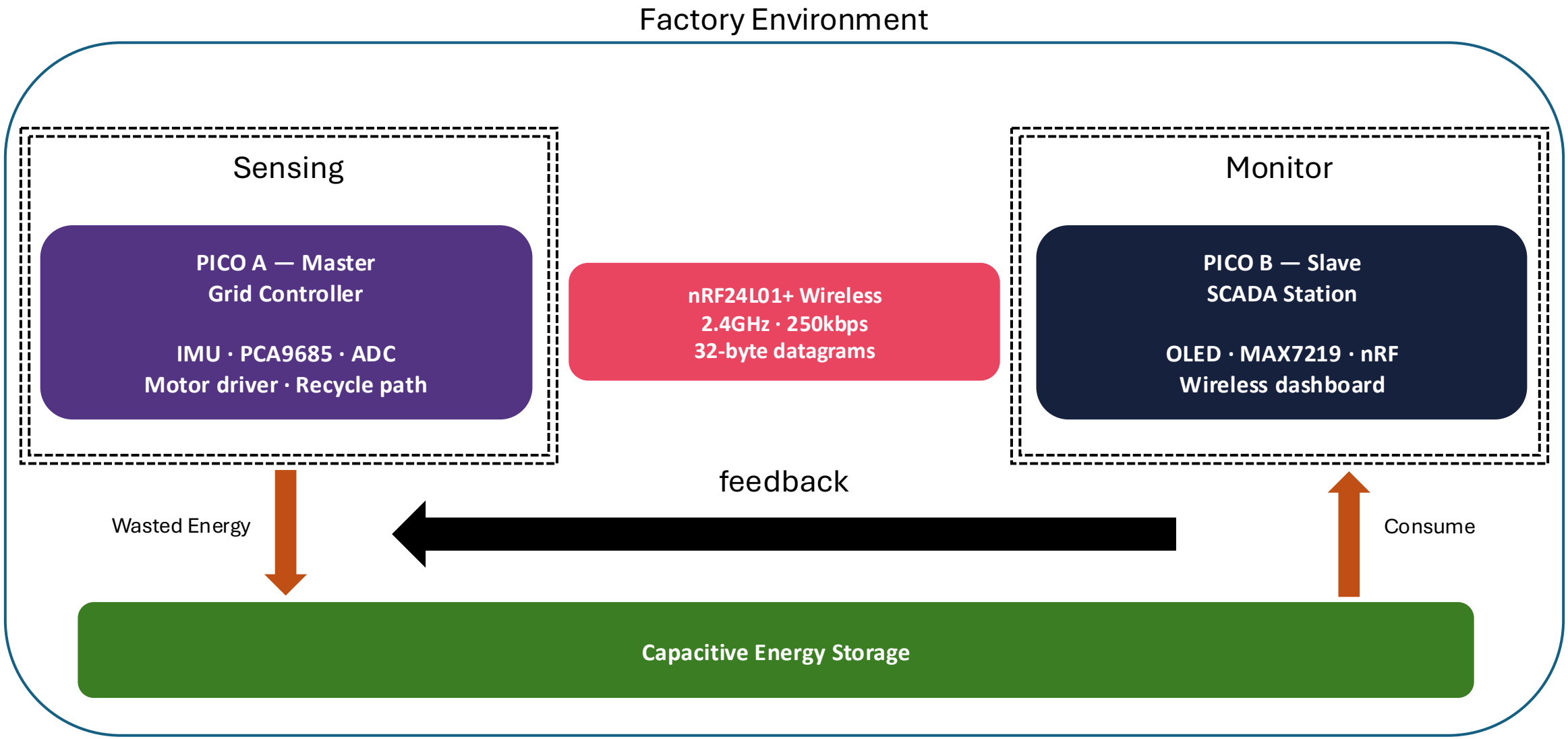
80% of small factories
priced out

Factories optimise how much energy they USE, but nobody captures the energy they've already LOST.

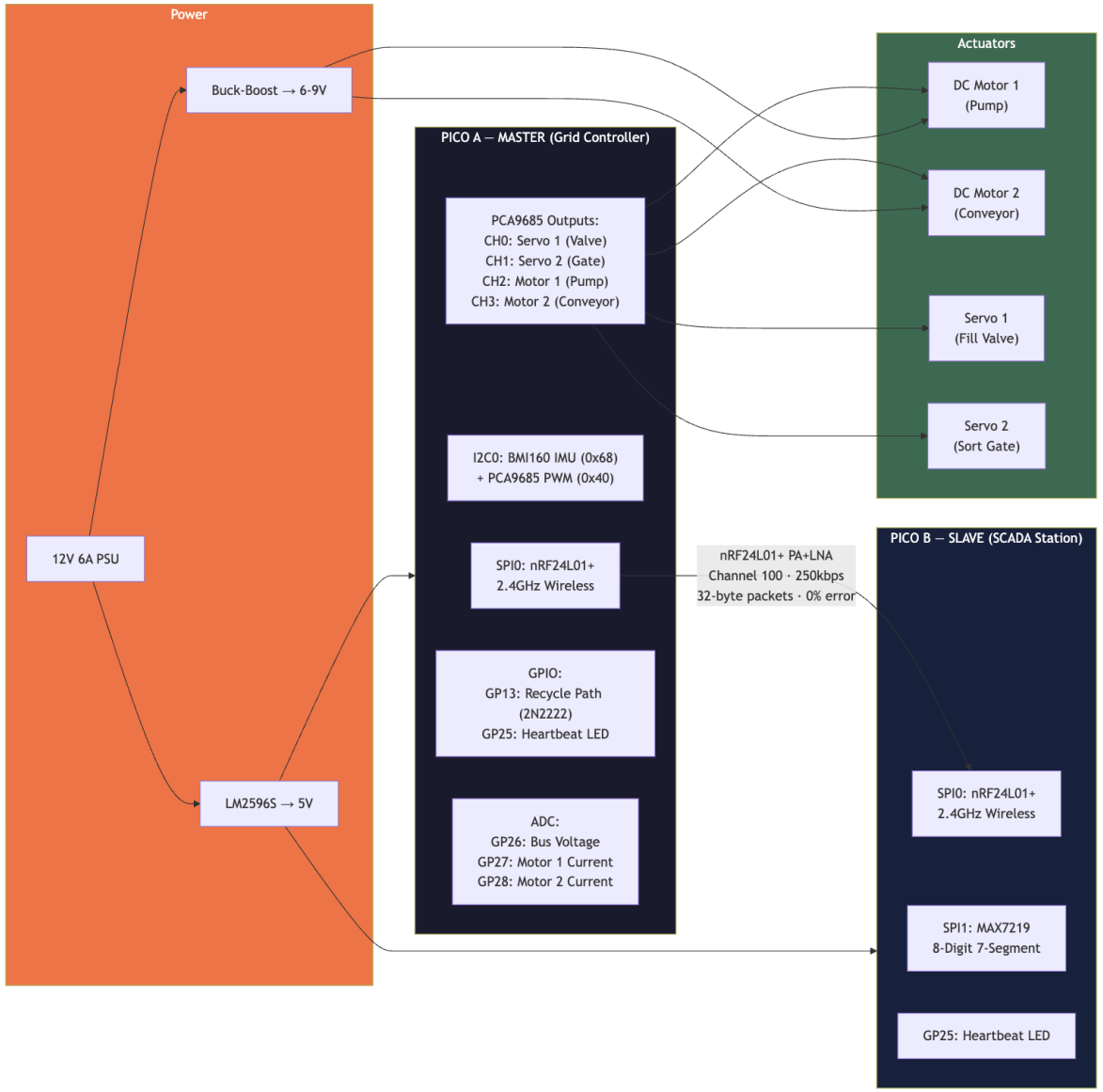
Braking, load transitions, idle cycles that energy becomes heat. In a car, regenerative braking recovers 30%. In a factory?

Nothing. Because regen systems cost £5,000+.

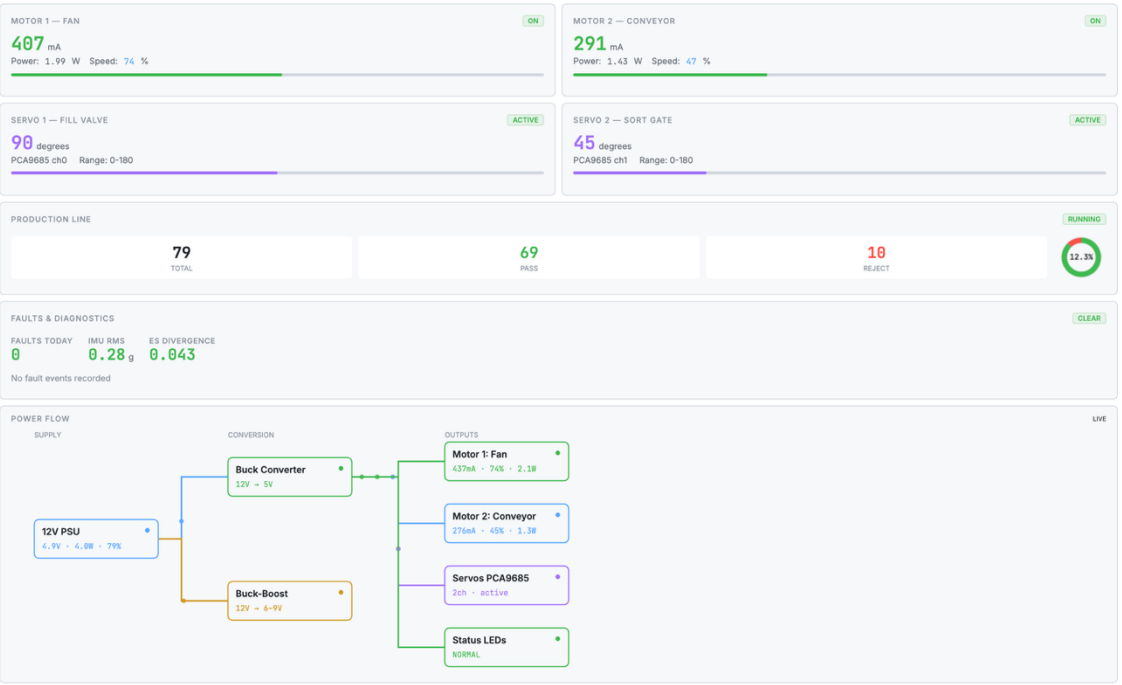
Our Solution



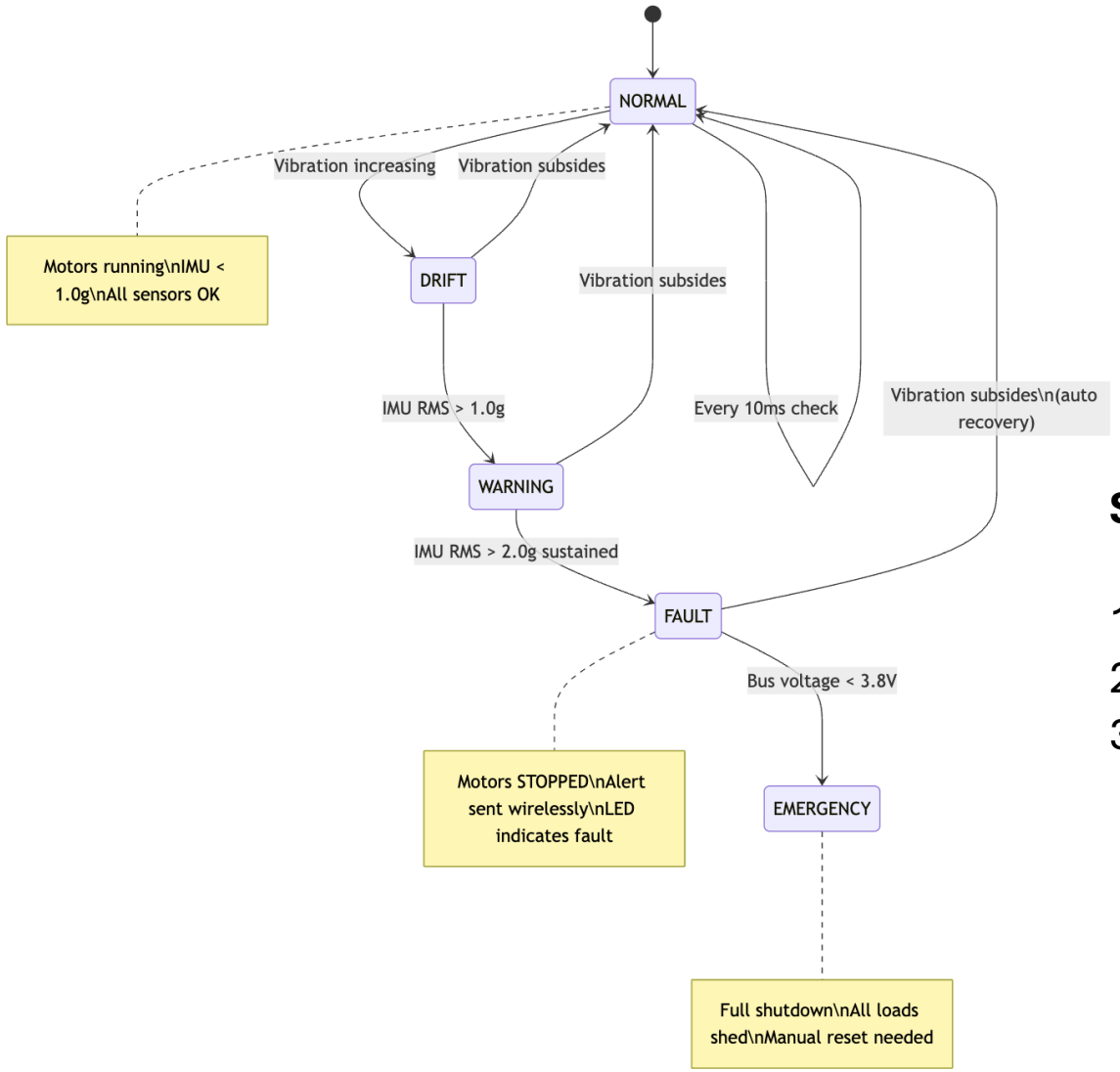
Systematic Architecture



Web Dashboard: `http://localhost:8080/`

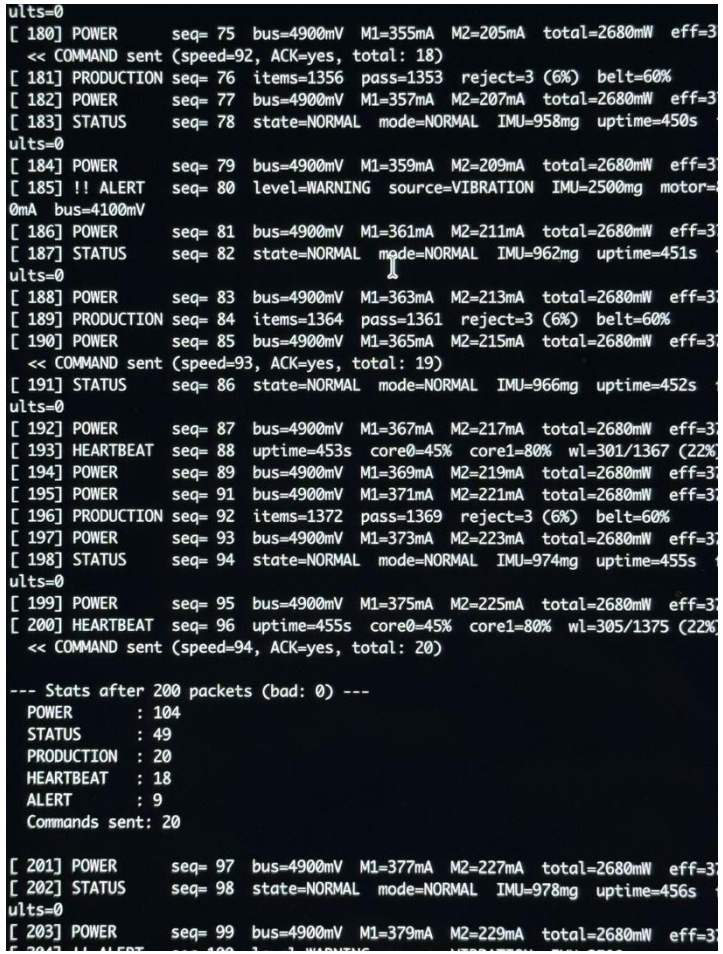


Technical Summary - Custom 6 type Datagram Protocol



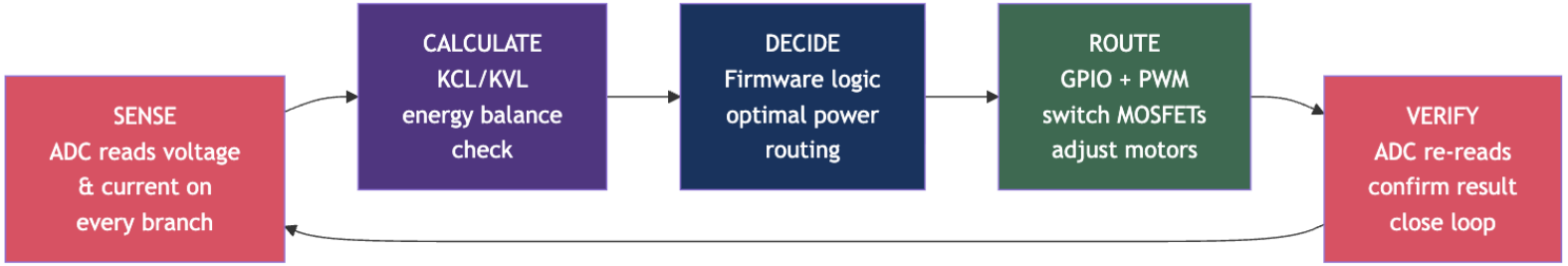
Strategies

1. Sequence Numbers
2. Redundant sends
3. Graceful degradation

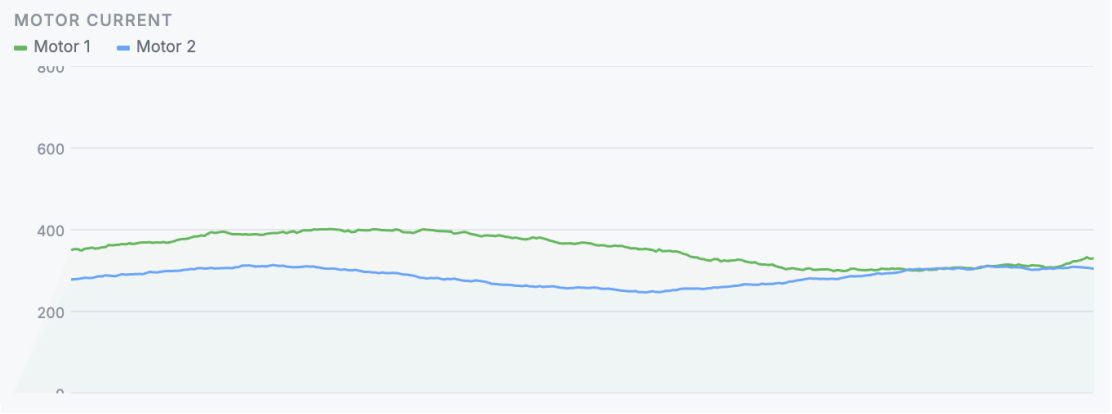


```
ults=0
[ 180] POWER      seq= 75  bus=4900mV  M1=355mA  M2=205mA  total=2680mW  eff=3
  << COMMAND sent (speed=92, ACK=yes, total: 18)
[ 181] PRODUCTION seq= 76  items=1356  pass=1353  reject=3 (6%)  belt=60%
[ 182] POWER      seq= 77  bus=4900mV  M1=357mA  M2=207mA  total=2680mW  eff=3
[ 183] STATUS     seq= 78  state=NORMAL  mode=NORMAL  IMU=958mg  uptime=450s
ults=0
[ 184] POWER      seq= 79  bus=4900mV  M1=359mA  M2=209mA  total=2680mW  eff=3
[ 185] !! ALERT   seq= 80  level=WARNING  source=VIBRATION  IMU=2500mg  motor=
0mA  bus=4100mV
[ 186] POWER      seq= 81  bus=4900mV  M1=361mA  M2=211mA  total=2680mW  eff=3
[ 187] STATUS     seq= 82  state=NORMAL  mode=NORMAL  IMU=962mg  uptime=451s
ults=0
[ 188] POWER      seq= 83  bus=4900mV  M1=363mA  M2=213mA  total=2680mW  eff=3
[ 189] PRODUCTION seq= 84  items=1364  pass=1361  reject=3 (6%)  belt=60%
[ 190] POWER      seq= 85  bus=4900mV  M1=365mA  M2=215mA  total=2680mW  eff=3
  << COMMAND sent (speed=93, ACK=yes, total: 19)
[ 191] STATUS     seq= 86  state=NORMAL  mode=NORMAL  IMU=966mg  uptime=452s
ults=0
[ 192] POWER      seq= 87  bus=4900mV  M1=367mA  M2=217mA  total=2680mW  eff=3
[ 193] HEARTBEAT  seq= 88  uptime=453s  core0=45%  core1=80%  wl=301/1367 (22%
[ 194] POWER      seq= 89  bus=4900mV  M1=369mA  M2=219mA  total=2680mW  eff=3
[ 195] POWER      seq= 91  bus=4900mV  M1=371mA  M2=221mA  total=2680mW  eff=3
[ 196] PRODUCTION seq= 92  items=1372  pass=1369  reject=3 (6%)  belt=60%
[ 197] POWER      seq= 93  bus=4900mV  M1=373mA  M2=223mA  total=2680mW  eff=3
[ 198] STATUS     seq= 94  state=NORMAL  mode=NORMAL  IMU=974mg  uptime=455s
ults=0
[ 199] POWER      seq= 95  bus=4900mV  M1=375mA  M2=225mA  total=2680mW  eff=3
[ 200] HEARTBEAT  seq= 96  uptime=455s  core0=45%  core1=80%  wl=305/1375 (22%
  << COMMAND sent (speed=94, ACK=yes, total: 20)

--- Stats after 200 packets (bad: 0) ---
  POWER       : 104
  STATUS      : 49
  PRODUCTION  : 20
  HEARTBEAT   : 18
  ALERT       : 9
  Commands sent: 20

[ 201] POWER      seq= 97  bus=4900mV  M1=377mA  M2=227mA  total=2680mW  eff=3
[ 202] STATUS     seq= 98  state=NORMAL  mode=NORMAL  IMU=978mg  uptime=456s
ults=0
[ 203] POWER      seq= 99  bus=4900mV  M1=379mA  M2=229mA  total=2680mW  eff=3
```

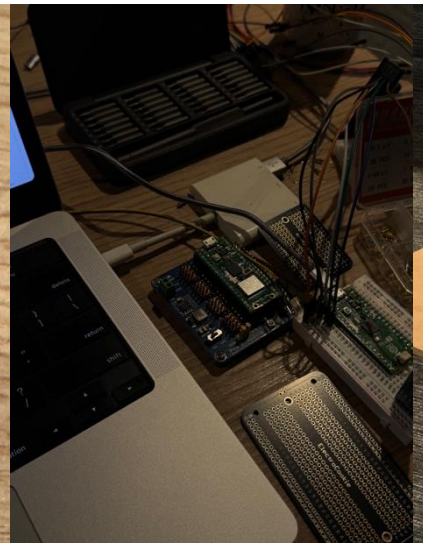
Applied Engineering: Closed-Loop Power Control

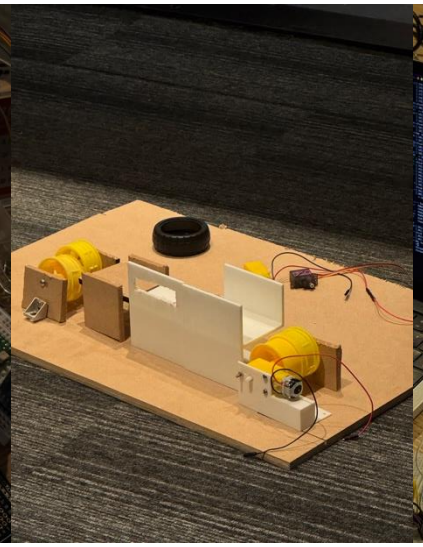


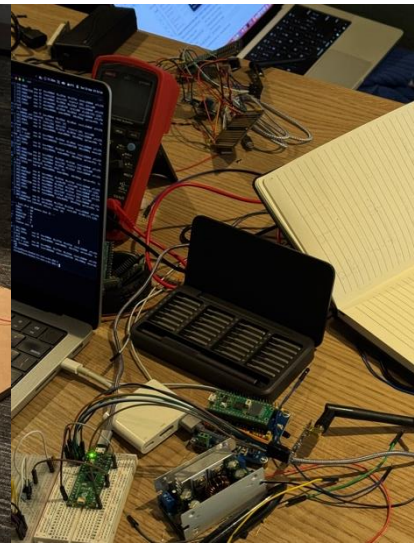
Core Theory	Equation	Outcome
Affinity Laws	$P \propto n^3$	20% slower = 49 less power
Current Sensing	$I = V_{sense}/R_{sense}$	Sense the wasted power

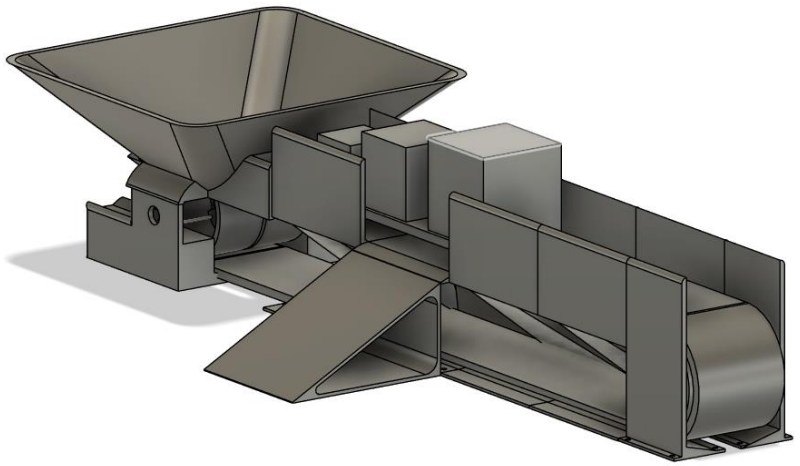


System Implementation Timeline

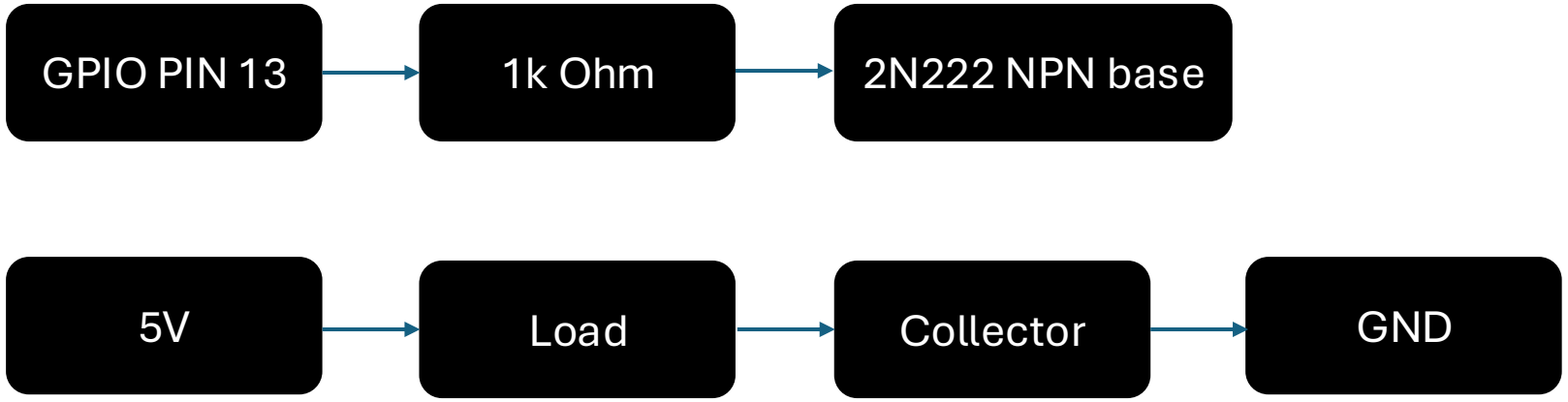








Failure Analysis



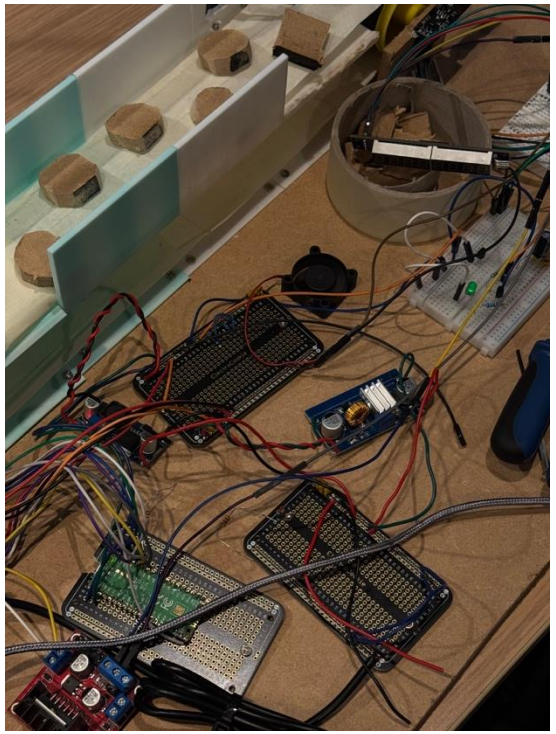
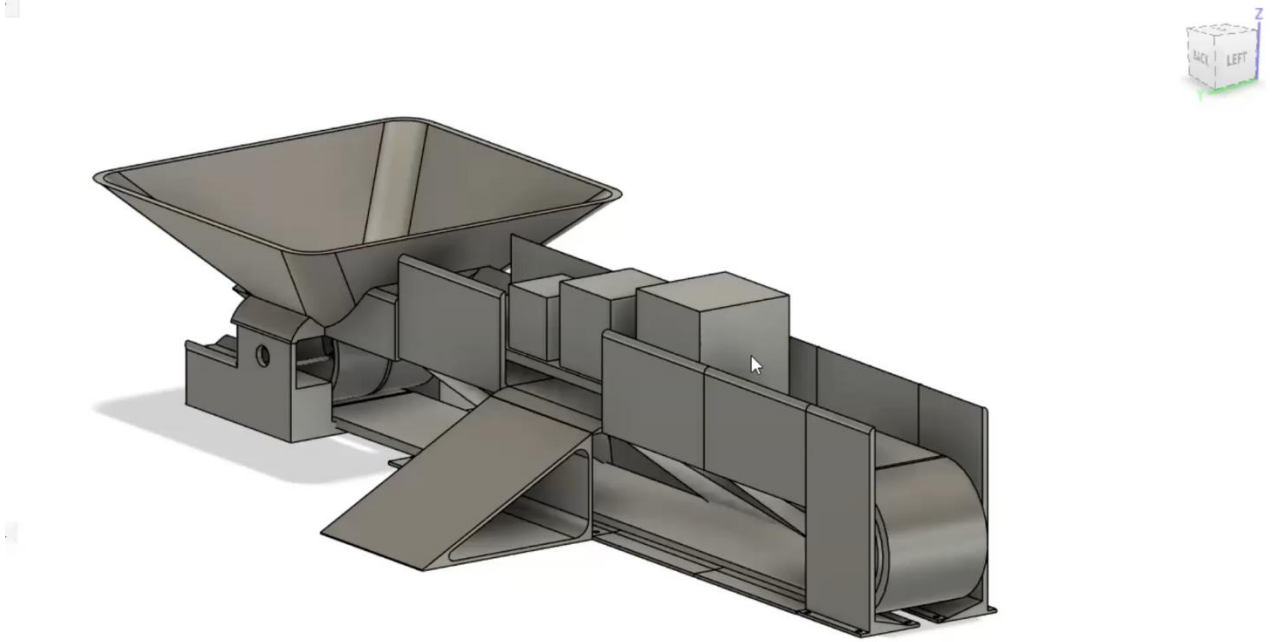
Max Collector Current: 880 mA

Max Gain Current: 330 mA $3.3\ mA\ \times 100 = 330\ mA$

We had motor for cooling and when the motor was turned off, there was reverse polarity voltage: 20 V – BURNT!

```
    ~~~~~~~~~~~~~~~~~~~~~~^^
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-p
ackages/mpremote/transport_serial.py", line 118, in close
    raise er
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-p
ackages/mpremote/transport_serial.py", line 111, in close
    self.serial.rts = False
    ^^^^^^^^^^^^^^^
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-p
ackages/serial/serialutil.py", line 463, in rts
    self._update_rts_state()
    ~~~~~~~~~~~~~~~~~~~~~~~^^
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-p
ackages/serial/serialposix.py", line 708, in _update_rts_state
    fcntl.ioctl(self.fd, TIOCMBIC, TIOCM_RTS_str)
    ~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
OSError: [Errno 6] Device not configured
doyungu@Doyuns-MacBook-Pro hack-a-bot-2026 %
doyungu@Doyuns-MacBook-Pro hack-a-bot-2026 % mpremote run src/slave-pico/tests/t
est_slave_full.py
mpremote: no device found
```

Thank you for Listening

All work is on GitHub including C source code!

https://github.com/doyun-gu/hack-a-bot-2026